STUDYHUB

CLASS 12 — MP BOARD

CHEMISTRY — SET 1

ORIGINAL SAMPLE QUESTION PAPER

Time: 3 Hours	Maximum Marks: 70
Medium: English	Set: 1

Note: This is an original StudyHub practice paper based on the MPBSE syllabus and examination pattern. It is not an official MPBSE paper.

GENERAL INSTRUCTIONS

1. All questions are compulsory.
2. Questions 1 to 5 are objective-type questions carrying a total of 28 marks.
3. Questions 6 to 12 carry 2 marks each. Answer within approximately 30 words.
4. Questions 13 to 16 carry 3 marks each. Answer within approximately 75 words.
5. Questions 17 to 20 carry 4 marks each. Answer within approximately 120 words.
6. Internal choices are provided from Question 6 to Question 20.
7. Write balanced chemical equations wherever required.

SECTION – A : OBJECTIVE TYPE QUESTIONS

Q.1 Choose the correct option. (1×6=6)

(i) The number of atoms present in a body-centred cubic unit cell is:
A. 1
B. 2
C. 4
D. 6

(ii) The colligative property most suitable for determining the molar mass of a protein is:
A. Elevation in boiling point
B. Depression in freezing point
C. Osmotic pressure
D. Relative lowering of vapour pressure

(iii) The oxidation state of chromium in $K_2Cr_2O_7$ is:
A. +2
B. +3
C. +6
D. +7

(iv) Which of the following is a strong electrolyte?
A. CH_3COOH
B. NH_4OH
C. HCl
D. H_2CO_3

(v) The functional group present in aldehydes is:
A. –OH
B. –COOH
C. –CHO
D. –CO–

(vi) Which vitamin is mainly associated with blood clotting?
A. Vitamin A
B. Vitamin C
C. Vitamin D
D. Vitamin K

Q.2 Fill in the blanks. (1×6=6)

(i) The concentration of a solution expressed as moles of solute per litre of solution is called ______.

(ii) The electrode at which reduction takes place is called the ______.

(iii) The rate constant of a first-order reaction has the unit ______.

(iv) The catalyst used in the Haber process is ______.

(v) The simplest aromatic hydrocarbon is ______.

(vi) The bond joining two monosaccharide units is called a ______ bond.

Q.3 Write True or False. (1×5=5)

(i) Molarity changes with temperature.

(ii) The standard electrode potential of a hydrogen electrode is taken as zero.

(iii) A catalyst changes the equilibrium constant of a reaction.

(iv) Primary alcohols can be oxidised to aldehydes.

(v) Proteins are polymers of amino acids.

Q.4 Match Column A with Column B. (1×6=6)

Column A	Column B
(i) Raoult's law	(a) –COOH
(ii) Faraday constant	(b) Vapour pressure
(iii) Carboxylic acid	(c) 96500 C mol^{-1}
(iv) Aldehyde	(d) –CHO
(v) Amine	(e) $–NH_2$
(vi) Nylon-6,6	(f) Polyamide

Q.5 Answer in one word/one sentence. (1×5=5)

(i) What is the coordination number of a face-centred cubic crystal?

(ii) What is the SI unit of conductivity?

(iii) What is the process of coating an iron object with zinc called?

(iv) Name the alcohol present in spirit used as a fuel.

(v) What is the monomer of polyethylene?

SECTION – B : VERY SHORT ANSWER QUESTIONS

Q.6 (2 Marks)
Define molality. How is it different from molarity?

OR
What is an ideal solution? Give one characteristic of an ideal solution.

Q.7 (2 Marks)
State Kohlrausch's law of independent migration of ions.

OR
What is meant by electrochemical cell? Write one example.

Q.8 (2 Marks)
Define order of a chemical reaction. How is it determined experimentally?

OR
What is activation energy? How does a catalyst affect it?

Q.9 (2 Marks)
Why do transition elements show variable oxidation states?

OR
Why are transition metal compounds generally coloured?

Q.10 (2 Marks)
What is lanthanide contraction? State one consequence of it.

OR
Why is NH_3 more basic than PH_3?

Q.11 (2 Marks)
Write the IUPAC name of CH_3CH_2CHO.

OR
What happens when ethanol is heated with concentrated H_2SO_4 at 443 K? Write the equation.

Q.12 (2 Marks)
What are essential amino acids?

OR
Differentiate between addition polymers and condensation polymers.

SECTION – C : SHORT ANSWER QUESTIONS

Q.13 (3 Marks)
Explain Henry's law for the solubility of a gas in a liquid. Write its mathematical form.

OR
A solution contains 5 g of glucose (M = 180 g mol^{-1}) dissolved in 95 g of water. Calculate the molality of the solution.

Q.14 (3 Marks)
Explain the construction and working of a Daniell cell with its cell notation.

OR
Calculate the standard cell potential for a cell having $E°_{cathode} = +0.34$ V and $E°_{anode} = -0.76$ V.

Q.15 (3 Marks)
Explain the mechanism of nucleophilic substitution reaction of an alkyl halide with aqueous KOH.

OR
Write chemical equations for: (i) Ethanol with sodium (ii) Ethanol with acidified potassium dichromate (iii) Phenol with bromine water.

Q.16 (3 Marks)
What are amines? Classify them into primary, secondary and tertiary amines with one example of each.

OR
Explain why aniline is less basic than ethylamine.

SECTION – D : LONG ANSWER QUESTIONS

Q.17 (4 Marks)
Derive the integrated rate equation for a first-order reaction and obtain the expression for its half-life.

OR
For a first-order reaction, the rate constant is 2.31×10^{-3} s^{-1}. Calculate its half-life.

Q.18 (4 Marks)
Explain the general characteristics of transition elements with reference to: (i) Variable oxidation states (ii) Formation of coloured ions (iii) Magnetic properties (iv) Complex formation.

OR
Explain the preparation and important properties of potassium permanganate.

Q.19 (4 Marks)
Explain the preparation of phenol from benzene diazonium chloride. Also write any two chemical reactions of phenol.

OR
Explain the preparation of aldehydes by oxidation of primary alcohols. Write the reaction with a suitable example.

Q.20 (4 Marks)
Explain the structure of glucose and discuss its important chemical properties.

OR
What are polymers? Explain addition polymerisation and condensation polymerisation with suitable examples.

MARKS VERIFICATION

Questions	Marks
Q.1	6
Q.2	6
Q.3	5
Q.4	6
Q.5	5
Q.6–Q.12	14
Q.13–Q.16	12
Q.17–Q.20	16
TOTAL	70

6 + 6 + 5 + 6 + 5 + 14 + 12 + 16 = 70 Marks ✓

SHORT ANSWER KEY

Q.1: B, C, C, C, C, D. Q.2: molarity; cathode; s^{-1}; iron; benzene; glycosidic. Q.3: True, True, False, True, True. Q.4: (i)-(b), (ii)-(c), (iii)-(a), (iv)-(d), (v)-(e), (vi)-(f). Q.5: 12; $S\ m^{-1}$; galvanisation; ethanol; ethene.

Q.13: Molality = moles of solute / kg of solvent. Q.14: $E^\circ cell = 1.10$ V. Q.17: $t_{1/2} = 0.693/k$. For Q.6–Q.20, award marks for correct formulae, equations, reasoning and calculations as applicable. Full solutions are not included.